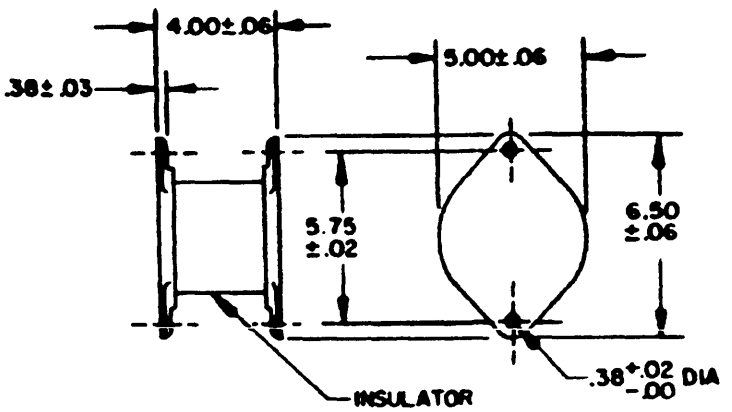
MILITARY SPECIFICATION SHEET
CAPACITORS, FIXED, MICA DIELECTRIC, STYLES CM85, CM86, AND CM88 1/

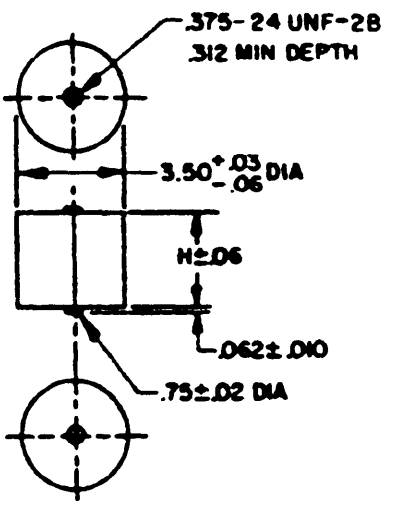
This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-5.

STYLE CM85 IS INACTIVE FOR NEW DESIGN AFTER THE DATE OF THIS SPECIFICATION.



STYLE CM85

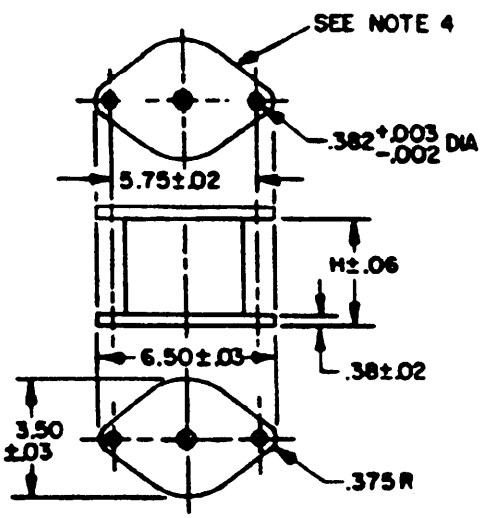


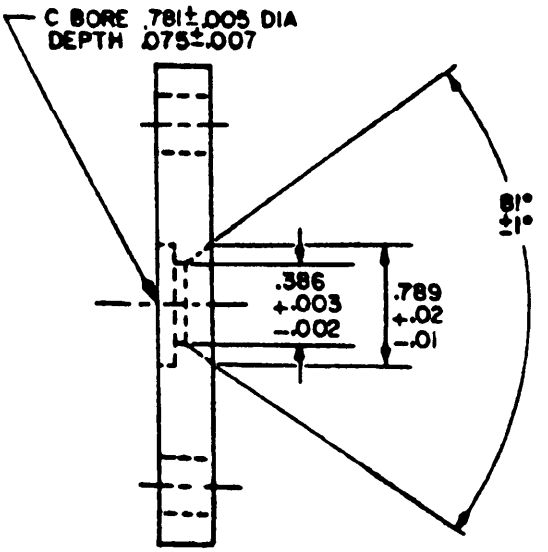
STYLE CM86

1/ CM87 deleted.

FSC 5910

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STYLE CM85

INCHES	MM
.002	.05
.003	.08
.010	.25
.02	.51
.03	.76
.06	1.52
.062	1.57
.312	7.92
.375	9.53
.38	9.65
.382	9.70
.386	9.80
.75	19.05
.781	19.84
.789	20.04
3.50	88.90
4.00	101.60
5.00	127.00
5.75	146.05
6.50	165.10

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.005(.13 mm) for three place decimals.
4. Brackets (flanges) shall be removable.
5. Hardware to be plated against corrosion.
6. Approximate weight CM85 is 5.5 pounds (2.49 kg).

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REQUIREMENTS:

Dimensions: See figure.

Construction: CM85 - Molded case.
CM86 and CM88 - Cast epoxy case.

Capacitance value: See table I.

Capacitance tolerance: ±5% (symbol J).

Peak working voltage: See table I.

Operating temperature range: See table I.

Symbol	Temperature range
M	-55 to +70° C
O	-55 to +125° C

Vibration grade: Grade 1 (10-55 Hz).

Characteristic: See table I.

Symbol	Temperature coefficient P/M/° C	Capacitance drift
B	Not rated	Not rated
D	-100 to +100	±(0.3 percent +0.1 pF)
E	-20 to +100	±(0.1 percent +0.1 pF)
F	0 to +70	±(0.05 percent +0.1 pF)

Dielectric withstanding voltage:
Method 301 of MIL-STD-202:
AC sine-wave potential rms equal to peak working voltage at 100 cycles or less for not less than 5 seconds.

Barometric pressure:
Method 105 of MIL-STD-202, condition B (50,000 ft).
Test potential: Capacitors rated above 12,000 volts - 45 percent of rated peak voltage; capacitors rated at 8,000 thru 12,000 volts - 70 percent of rated peak voltage; capacitors rated at 5,000 volts or less - 150 percent of rated peak voltage.

Insulation resistance:
Method 302 of MIL-STD-202, condition A.
At 25° C:
(CM85) - 7,500 megohms minimum.
(CM86 and CM88) - 10,000 megohms minimum.
At 70° C:
(CM85) - 0 to 5,000 pF--2,000 megohms minimum; greater than 5,000 pF--10 megohm microfarad.
At 125° C:
(CM86 and CM88) - 0 to 5,000 pF--3 megohms minimum; greater than 5,000 pF--15 megohm microfarads.

Dissipation factor, maximum - See MIL-C-5.

Capacitance:
Method 305 of MIL-STD-202.

Current rating:
Passage of rated current shall not cause the temperature of the capacitor body's external surface to rise above the ambient temperature by more than 15° C for style CM85 and not more than 40° C for styles CM86 and CM88.

Vibration:
Method 201 of MIL-STD-202.

Temperature cycling:
Method 102 of MIL-STD-202, condition D, except the high temperature shall be as follows:
Symbol M - +70° C
Symbol O - +125° C.

Immersion cycling:
Method 104 of MIL-STD-202, condition B.
Dielectric withstanding voltage - Meet initial requirement.
Insulation resistance - 3,000 megohms minimum.
Dissipation factor - Not greater than 150 percent of initial value.
Capacitance change - Shall not exceed ±4 percent of nominal value or 1 pF, whichever is greater.

Shock (specified pulse): Method 213 of MIL-STD-202, condition I (100G(pk)).

Moisture resistance:
Method 106 of MIL-STD-202:
Insulation resistance - At least 30 percent of initial requirement.
Dissipation factor - Not greater than 150 percent of initial value.

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Capacitance change - Shall not exceed ±5 percent of nominal value or 1 pF, whichever is greater, for characteristic B; ±2 percent of nominal value or 1 pF, whichever is greater, for characteristics D, E, and F.

Temperature coefficient and capacitance drift: Within the limits specified for the applicable characteristic.

Low temperature storage and life:

48 hours at -55 $^{+0}_{-2}$ C immediately followed by qualification or performance test, as applicable.

Qualification test - 1,000 hours at the applicable high test temperature with rated current at 1 MHz.

Dielectric withstanding voltage - Initial requirement.

Insulation resistance - Initial requirement.

Dissipation factor - Not greater than 150 percent of initial requirement.

Capacitance change - Shall not exceed ±8 percent of nominal value or 1 pF, whichever is greater, for characteristic B; ±3 percent of nominal value or 1 pF, whichever is greater, for characteristics D, E, and F.

Performance check - 250 hours at conditions specified for qualification test.

Dielectric withstanding voltage - Initial requirement.

Insulation resistance - Initial requirement.

Dissipation factor - Not greater than 150 percent of initial requirement.

Capacitance change - Shall not exceed ±5 percent of nominal value or 1 pF, whichever is greater, for characteristic B; ±2 percent of nominal value or 1 pF, whichever is greater, for characteristics D, E, and F.

Continuation test - Units that have been subjected to the 250 hour test shall be continued for an additional 750 hours and meet the requirements for the qualification test. (This test not applicable to styles CM86 and CM88.)

Qualification inspection:

Qualification inspection (Group E) is required.

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TABLE I. Styles CM85, CM86, and CM88.

Type designation 1/	Capacitance value (pF)	Peak working voltage (volts)	Rated current at frequencies (rms amps) 3.0 mHz	1.0 mHz	0.3 mHz	0.1 mHz	Characteristic
CM85-U470JM1	47	20,000	5.1	2.2	0.47	0.12	B, D, F
CM85-U510JM1	51	20,000	5.1	2.4	0.51	0.15	
CM85-U560JM1	56	20,000	5.6	2.4	0.56	0.18	
CM85-U620JM1	62	20,000	5.6	2.7	0.68	0.20	
CM85-U680JM1	68	20,000	5.6	2.7	0.75	0.24	
CM85-U750JM1	75	20,000	6.2	3.0	0.91	0.27	
CM85-U820JM1	82	20,000	6.2	3.0	1.0	0.33	
CM85-U910JM1	91	20,000	6.8	3.3	1.1	0.39	
CM85-U101JM1	100	20,000	6.8	3.6	1.3	0.47	
CM85-U111JM1	110	20,000	7.5	3.9	1.5	0.51	
CM85-U121JM1	120	20,000	7.5	4.3	1.6	0.62	
CM85-U131JM1	130	20,000	7.5	4.3	1.8	0.68	
CM85-U151JM1	150	20,000	8.2	4.7	2.0	0.82	
CM85-U161JM1	160	20,000	8.2	5.1	2.2	0.91	
CM85-U181JM1	180	20,000	9.1	5.1	2.4	1.0	
CM85-U201JM1	200	20,000	9.1	5.6	2.7	1.1	
CM85-U221JM1	220	20,000	9.1	6.2	3.0	1.2	
CM85-U241JM1	240	20,000	10	6.2	3.6	1.3	
CM85-U271JM1	270	20,000	10	6.8	3.6	1.5	
CM85-U301JM1	300	20,000	10	6.8	3.9	1.6	
CM85-U331JM1	330	20,000	11	7.5	3.9	1.6	
CM85-U361JM1	360	20,000	11	7.5	3.9	1.8	
CM85-U391JM1	390	20,000	11	8.2	4.3	2.0	
CM85-U431JM1	430	20,000	12	8.2	4.7	2.2	
CM85-U471JM1	470	20,000	12	9.1	4.7	2.2	
CM85-U511JM1	510	20,000	12	9.1	5.1	2.4	
CM85-U561JM1	560	20,000	13	10	5.6	2.7	
CM85-U621JM1	620	20,000	13	11	6.2	2.7	
CM85-U681JM1	680	20,000	13	11	6.2	2.7	
CM85-U751JM1	750	20,000	13	12	6.8	3.0	
CM85-U821JM1	820	20,000	15	12	6.8	3.3	
CM85-U911JM1	910	20,000	15	13	7.5	3.6	
CM85-U102JM1	1000	20,000	15	13	7.5	3.6	
CM85-U112JM1	1100	20,000	16	15	8.2	3.9	
CM85-T122JM1	1200	15,000	16	15	8.2	4.3	
CM85-T132JM1	1300	15,000	16	15	9.1	4.3	
CM85-T152JM1	1500	15,000	16	16	9.1	4.7	
CM85-T162JM1	1600	15,000	18	18	10	4.7	
CM85-T182JM1	1800	15,000	18	18	11	5.1	
CM85-T202JM1	2000	15,000	18	20	11	5.6	
CM85-S222JM1	2200	12,000	20	20	12	5.6	
CM85-S242JM1	2400	12,000	20	22	13	6.2	
CM85-S272JM1	2700	12,000	20	22	13	6.2	
CM85-S302JM1	3000	12,000	20	24	13	6.8	
CM85-S332JM1	3300	12,000	22	24	15	6.8	
CM85-S362JM1	3600	12,000	22	24	15	7.5	
CM85-S392JM1	3900	12,000	22	27	16	7.5	
CM85-S432JM1	4300	12,000	22	27	16	8.2	
CM85-R472JM1	4700	10,000	22	27	18	8.2	
CM85-R512JM1	5100	10,000	24	30	18	9.1	
CM85-R562JM1	5600	10,000	24	33	20	9.1	
CM85-R622JM1	6200	10,000	24	33	20	10	B, D, F

See footnotes at end of table.

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TABLE I. Styles CM85, CM86, and CM88. - Continued

Type designation 1/	Capacitance value (pF)	Peak working voltage (volts)	Rated current at frequencies (rms amps) 3.0 mHz	1.0 mHz	0.3 mHz	0.1 mHz	Characteristic
CM85-R682JM1	6800	10,000	24	33	20	10	B, D, F
CM85-R752JM1	7500	10,000	24	36	22	11	
CM85-R822JM1	8200	10,000	24	36	22	11	
CM85-Q912JM1	9100	8,000	24	39	24	12	
CM85-Q103JM1	10,000	8,000	24	39	24	12	
CM85-Q113JM1	11,000	8,000	24	39	27	13	
CM85-N123JM1	12,000	5,000	24	43	27	13	
CM85-N133JM1	13,000	5,000	24	43	27	13	
CM85-N153JM1	15,000	5,000	24	43	27	15	
CM85-N163JM1	16,000	5,000	24	47	30	15	
CM85-N183JM1	18,000	5,000	24	47	30	16	
CM85-N203JM1	20,000	5,000	24	47	30	18	
CM85-N223JM1	22,000	5,000	24	47	30	18	
CM85-N243JM1	24,000	5,000	24	51	33	18	
CM85-L273JM1	27,000	3,000	24	51	33	18	
CM85-L303JM1	30,000	3,000	24	51	33	20	
CM85-L333JM1	33,000	3,000	24	51	33	20	
CM85-L363JM1	36,000	3,000	24	51	36	22	
CM85-L393JM1	39,000	3,000	24	51	36	22	
CM85-L433JM1	43,000	3,000	24	51	36	22	
CM85-L473JM1	47,000	3,000	24	51	36	22	
CM85-L513JM1	51,000	3,000	24	51	36	24	
CM85-L563JM1	56,000	3,000	24	51	36	24	
CM85-L623JM1	62,000	3,000	24	51	39	24	
CM85-L683JM1	68,000	3,000	24	51	39	24	
CM85-J753JM1	75,000	2,000	24	51	39	24	
CM85-J823JM1	82,000	2,000	24	51	39	24	
CM85-J913JM1	91,000	2,000	24	51	39	24	
CM85-1104JM1	100,000	2,000	24	51	39	24	B, D, F

Type designation 2/	Capacitance (pF)	Rated peak voltage (volts)	Case height H (inches)	Rated current at frequencies (rms amps) 3.0 mHz	1.0 mHz	0.3 mHz	0.1 mHz	Characteristic
CM8--U470JO1	47	20,000	2.50	6.1	2.2	.47	.12	D, E
CM8--U510JO1	51	20,000	2.50	5.1	2.5	.51	.15	
CM8--U560JO1	56	20,000	2.50	5.6	2.4	.56	.18	
CM8--U620JO1	62	20,000	2.50	5.6	2.7	.68	.20	
CM8--U680JO1	68	20,000	2.50	5.6	2.7	.75	.24	
CM8--U750JO1	75	20,000	2.50	6.2	3.0	.91	.27	
CM8--U820JO1	82	20,000	2.50	6.2	3.0	1.0	.33	
CM8--U910JO1	91	20,000	2.50	6.8	3.3	1.1	.39	
CM8--U101JO1	100	20,000	2.50	6.8	3.6	1.3	.47	
CM8--U111JO1	110	20,000	2.50	7.5	3.9	1.5	.51	
CM8--U121JO1	120	20,000	2.50	7.5	4.3	1.6	.62	
CM8--U131JO1	130	20,000	2.50	7.5	4.3	1.8	.68	
CM8--U151JO1	150	20,000	2.50	8.2	4.7	2.0	.82	
CM8--U161JO1	160	20,000	2.50	8.2	5.1	2.2	.91	
CM8--U181JO1	180	20,000	2.50	9.1	5.1	2.4	1.0	
CM8--U201JO1	200	20,000	2.50	9.1	5.6	2.7	1.1	
CM8--U221JO1	220	20,000	2.50	9.1	6.2	3.0	1.2	
CM8--U241JO1	240	20,000	2.50	10	6.2	3.6	1.3	
CM8--U251JO1	250	20,000	2.50	10	6.2	3.6	1.3	D, E

See footnotes at end of table.

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TABLE I. Styles CM85, CM86, and CM88. - Continued

Type designation 2/	Capacitance (pF)	Rated peak voltage (volts)	Case height H (inches)	Rated current at frequencies (rms amps) 3.0 mHz	1.0 mHz	0.3 mHz	0.1 mHz	Characteristic
CM8--U271JO1	270	20, 000	2.50	10	6.8	3.8	1.6	D, E
CM8--U301JO1	300	20, 000	2.50	10	6.8	3.8	1.6	
CM8--U331JO1	330	20, 000	2.50	11	7.5	3.9	1.8	
CM8--U361JO1	360	20, 000	2.50	11	7.5	3.9	1.8	
CM8--U391JO1	390	20, 000	2.50	11	8.2	4.3	2.0	
CM8--U431JO1	430	20, 000	2.50	12	8.2	4.7	2.2	
CM8--U471JO1	470	20, 000	2.50	12	9.1	4.7	2.2	
CM8--U511JO1	510	20, 000	2.50	12	9.1	5.1	2.4	
CM8--U561JO1	560	20, 000	2.50	13	10	5.6	2.7	
CM8--U621JO1	620	20, 000	2.50	13	11	6.2	2.7	
CM8--U681JO1	680	20, 000	2.75	13	11	6.2	2.7	
CM8--U751JO1	750	20, 000	2.75	13	12	6.8	3.0	
CM8--U821JO1	820	20, 000	2.75	15	12	6.8	3.3	
CM8--U911JO1	910	20, 000	2.75	15	13	7.5	3.6	
CM8--U102JO1	1000	20, 000	3.00	15	13	7.5	3.6	
CM8--U112JO1	1100	20, 000	3.00	16	15	8.2	3.9	
CM8--T122JO1	1200	15, 000	2.50	16	15	8.2	4.3	
CM8--T132JO1	1300	15, 000	2.50	16	15	9.1	4.3	
CM8--T152JO1	1500	15, 000	2.75	16	16	9.1	4.7	
CM8--T162JO1	1600	15, 000	2.75	18	18	10	4.7	
CM8--T182JO1	1800	15, 000	3.00	18	18	11	5.1	
CM8--T202JO1	2000	15, 000	3.00	18	20	11	5.6	
CM8--S222JO1	2200	12, 000	2.50	20	20	12	5.6	
CM8--S242JO1	2400	12, 000	2.75	20	22	13	6.2	
CM8--S272JO1	2700	12, 000	2.75	20	22	13	6.2	
CM8--S302JO1	3000	12, 000	3.00	20	24	13	6.8	
CM8--S332JO1	3300	12, 000	3.00	22	24	15	6.8	
CM8--S362JO1	3600	12, 000	3.00	22	24	15	7.5	
CM8--S392JO1	3900	12, 000	3.00	22	27	16	7.5	
CM8--S402JO1	4000	12, 000	3.25	22	27	16	7.5	
CM8--S432JO1	4300	12, 000	3.25	22	27	16	8.2	
CM8--R472JO1	4700	10, 000	2.75	22	27	18	8.2	
CM8--R512JO1	5100	10, 000	2.75	24	30	18	9.1	
CM8--R562JO1	5600	10, 000	3.00	24	33	20	9.1	
CM8--R622JO1	6200	10, 000	3.00	24	33	20	10	
CM8--R682JO1	6800	10, 000	3.00	24	33	20	10	
CM8--R752JO1	7500	10, 000	3.25	24	36	22	11	
CM8--R822JO1	8200	10, 000	3.25	24	36	22	11	
CM8--Q912JO1	9100	8, 000	3.25	24	39	24	12	
CM8--Q103JO1	10, 000	8, 000	3.25	24	39	24	12	
CM8--Q113JO1	11, 000	8, 000	3.25	24	39	27	13	
CM8--N123JO1	12, 000	5, 000	2.50	24	43	27	13	
CM8--N133JO1	13, 000	5, 000	2.50	24	43	27	13	
CM8--N153JO1	15, 000	5, 000	2.50	24	43	27	15	
CM8--N163JO1	16, 000	5, 000	2.50	24	47	30	15	
CM8--N183JO1	18, 000	5, 000	2.75	24	47	30	16	
CM8--N203JO1	20, 000	5, 000	2.75	24	47	30	18	
CM8--N223JO1	22, 000	5, 000	3.00	24	47	30	18	
CM8--N243JO1	24, 000	5, 000	3.25	24	51	33	18	
CM8--L273JO1	27, 000	3, 000	2.50	24	51	33	18	
CM8--L303JO1	30, 000	3, 000	2.50	24	51	33	20	
CM8--L333JO1	33, 000	3, 000	2.50	24	51	33	20	
CM8--L363JO1	36, 000	3, 000	2.50	24	51	36	22	
CM8--L393JO1	39, 000	3, 000	2.50	24	51	36	22	
CM8--L433JO1	43, 000	3, 000	2.50	24	51	36	22	
CM8--L473JO1	47, 000	3, 000	2.50	24	51	36	22	
CM8--L513JO1	51, 000	3, 000	2.75	24	51	36	24	
CM8--L563JO1	56, 000	3, 000	2.75	24	51	36	24	
CM8--L623JO1	62, 000	3, 000	3.00	24	51	39	24	
CM8--L683JO1	68, 000	3, 000	3.00	24	51	39	24	
CM8--J753JO1	75, 000	2, 000	2.50	24	51	39	24	
CM8--J823JO1	82, 000	2, 000	2.50	24	51	39	24	
CM8--J913JO1	91, 000	2, 000	2.75	24	51	39	24	
CM8--J104JO1	100, 000	2, 000	2.75	24	51	39	24	D, E

1/ For style CM85, the complete type designation will include an additional symbol to indicate characteristic.
2/ For styles CM86 and CM88, the complete type designation will include additional symbols to indicate style and characteristic.

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Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - EL
Navy - AS, OS, EC, SH
Air Force - 11, 17, 80
DSA - ES

User activities:
Army -
Navy - MC, CG
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

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WASHINGTON, D. C. 20360

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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